

LUCIDMSK CME PROGRAM

MODULE 05 — POST-TEST

Hip MRI & THA: AVN Staging, FAI & Arthroplasty Complications

Hip · 10 Questions · 1.0 AMA PRA Cat 1 Credit(s)[™]

Read each question carefully and select the single best answer. A score of $\geq 70\%$ (7/10 correct) is required to claim CME credit. The complete answer key with detailed explanations begins after Question 10.

POST-TEST QUESTIONS

1

MRI shows "double line sign" at the anterosuperior femoral head; normal radiograph. Ficat-Arlet stage:

- A Stage I
- B Stage IIA
- C Stage IIB
- D Stage III

2

Alpha angle measures 68° at the anterosuperior position on radial oblique MRI. This indicates:

- A Normal morphology
- B Cam morphology (FAI)
- C Pincer morphology
- D Hip dysplasia

3

MARS MRI shows large cystic periarticular fluid collection, thin wall, no solid component around MoM THA. Anderson type:

- A Type 1
- B Type 2A (solid)
- C Type 2B (cystic)
- D Type 4 (destructive)

4

Post-THA periprosthetic fracture at stem tip with stem subsidence and lucent lines. Vancouver classification:

- A Type AL
- B Type B1
- C Type B2
- D Type C

5 LCEA measures 22° on standing AP pelvis. Most consistent with:

- A Normal coverage
- B Pincer FAI (overcoverage)
- C Hip dysplasia
- D Normal variant

6 MARS MRI shows mixed solid + cystic ALTR around a large-head metal-on-polyethylene THA. Anderson type:

- A Type 1 — monitor
- B Type 2A — MRI in 6 months
- C Type 3 mixed — early revision
- D Type 4 — emergency surgery

7 Brooker Grade IV HO is present 9 months post-THA. Next step:

- A Immediate surgical excision
- B Delayed excision until bone scan quiescence + prophylaxis for contralateral THA
- C Radiation alone
- D Observation only indefinitely

8 The crescent sign on MRI indicates which ARCO stage?

- A Stage 1 (pre-radiographic)
- B Stage 2 (sclerosis/cysts)
- C Stage 3 (subchondral fracture)
- D Stage 4 (established OA)

9 Which THA complication can progress to catastrophic head-neck dissociation?

A	Particle disease / osteolysis
B	Trunnionosis (taper corrosion)
C	Iliopsoas impingement
D	Heterotopic ossification

10 Vancouver Type C periprosthetic fracture management:

A	Always requires stem revision
B	ORIF without stem revision — fracture distal to stem tip
C	Conservative cast/brace
D	Proximal femoral replacement

ANSWER KEY & EXPLANATIONS

Q	1	2	3	4	5	6	7	8	9	10
ANS	A	B	C	C	C	C	B	C	B	B

Q1	MRI shows "double line sign" at the anterosuperior femoral head; normal radiograph. Ficat-Arlet stage:	Answer: A
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✓ **Stage I**

B Stage IIA

C Stage IIB

D Stage III

EXPLANATION Ficat Stage I = positive MRI (double line sign on T2) with NORMAL radiograph. Stage IIA requires X-ray sclerosis/cysts. Stage IIB = crescent sign. Stage III = femoral head collapse.

Q2	Alpha angle measures 68° at the anterosuperior position on radial oblique MRI. This indicates:	Answer: B
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A Normal morphology

✓ **Cam morphology (FAI)**

C Pincer morphology

D Hip dysplasia

EXPLANATION Alpha angle >55° = cam morphology. Normal <55°. 68° is significantly elevated, confirming cam FAI at the anterosuperior head-neck junction.

Q3	MARS MRI shows large cystic periarticular fluid collection, thin wall, no solid component around MoM THA. Anderson type:	Answer: C
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A Type 1

B Type 2A (solid)

✓ **Type 2B (cystic)**

D Type 4 (destructive)

EXPLANATION Anderson Type 2B = cystic/fluid-dominant pseudotumor with thin wall, no solid component. Type 2A = solid mass. Type 1 = synovitis without discrete mass. Type 4 = bone/muscle destruction.

Q4	Post-THA periprosthetic fracture at stem tip with stem subsidence and lucent lines. Vancouver classification:	Answer: C
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A Type AL

B Type B1

✓ **Type B2**

D Type C

EXPLANATION Vancouver B2 = fracture at/around stem tip with UNSTABLE stem (subsidence + lucent lines = loosening). B1 = same location with stable stem → ORIF. C = distal to stem. AL = lesser trochanter.

Q5 LCEA measures 22° on standing AP pelvis. Most consistent with:

Answer: C

A Normal coverage

B Pincer FAI (overcoverage)

✓ **Hip dysplasia**

D Normal variant

EXPLANATION LCEA <25° = hip dysplasia (inadequate lateral coverage). Normal = 25–40°. >40° = overcoverage/pincer. Dysplastic hips are at high risk for labral tears and early OA.

Q6 MARS MRI shows mixed solid + cystic ALTR around a large-head metal-on-polyethylene THA. Anderson type:

Answer: C

A Type 1 — monitor

B Type 2A — MRI in 6 months

✓ **Type 3 mixed — early revision**

D Type 4 — emergency surgery

EXPLANATION Type 3 = mixed solid + cystic ALTR. Large-head MoP hips can develop ALTR via trunnion corrosion. Mixed Type 3 lesions with aggressive features warrant early surgical revision discussion.

Q7 Brooker Grade IV HO is present 9 months post-THA. Next step:

Answer: B

A Immediate surgical excision

✓ **Delayed excision until bone scan quiescence + prophylaxis for contralateral THA**

C Radiation alone

D Observation only indefinitely

EXPLANATION Grade IV HO excision is delayed until bone maturity (bone scan quiescence, ~12–18 months). Premature excision risks recurrence. Indomethacin or local radiation prophylaxis for contralateral THA.

Q8 The crescent sign on MRI indicates which ARCO stage?

Answer: C

A Stage 1 (pre-radiographic)

B Stage 2 (sclerosis/cysts)

✓ **Stage 3 (subchondral fracture)**

D Stage 4 (established OA)

EXPLANATION Crescent sign = curvilinear T2 fluid signal (MRI) or lucency (X-ray) = subchondral fracture = ARCO Stage 3. This is the LAST opportunity for joint-preserving surgery before collapse.

Q9 Which THA complication can progress to catastrophic head-neck dissociation?

Answer: B

A Particle disease / osteolysis

✓ **Trunnionosis (taper corrosion)**

C Iliopsoas impingement

D Heterotopic ossification

EXPLANATION Trunnionosis = progressive corrosion at the femoral head-neck taper junction. If unchecked, it can cause complete head-neck dissociation. Occurs in MoM and large-head MoP bearings.

Q10 Vancouver Type C periprosthetic fracture management:

Answer: B

A Always requires stem revision

✓ **ORIF without stem revision — fracture distal to stem tip**

C Conservative cast/brace

D Proximal femoral replacement

EXPLANATION Vancouver C = fracture DISTAL to stem tip. The stem is uninvolved. Standard ORIF (plate ± cerclage) without stem revision — even if stem shows some loosening. Key distinction from B2.